

Claudio Perinuzzi

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TECHNICAL SKILLS

Healthcare IT: HL7, FHIR, DICOM, MWL, PACS, Mirth Connect, EMR Integration (Epic, Cerner, Meditech, eCW)
Programming/Data: Ruby, Python, JavaScript, Java, C/C++, SQL (PostgreSQL, MySQL), HTML/CSS, JSON, XML
Frameworks: Ruby on Rails, React, Streamlit, FastAPI, SQLAlchemy, Pandas, NumPy, PyTorch, JUnit, UnitTest
Tools/OS: Linux, Docker, VMware, Ansible, Git/GitHub, Bash Scripting, Jira, Asana, AWS (EC2), Maven, Conda

EXPERIENCE

Software Integration Engineer

October 2025 - Present

STERIS

Hauppauge, NY

- Integrate, test, and deploy enterprise healthcare software solutions within hospital enterprise networks, including configuring virtualized environments and consulting on complex IT architectures.
- Manage full-lifecycle software integration projects, collaborating with hospital IT teams, internal stakeholders, and third-party vendors to ensure seamless data exchange and system interoperability.
- Configure inbound clinical data pipelines, processing HL7 SIU scheduling messages and DICOM C-FIND (MWL) queries to route patient and surgical data to a PostgreSQL database and downstream OR devices.
- Orchestrate outbound clinical export pipelines, ensuring captured OR media is archived to a central server and routed via enterprise NAS dropoffs, HL7 ORU messages to EMRs, or DICOM C-STORE to PACS.
- Manage high-stakes EMR cutovers and system go-lives, providing continuous monitoring and Tier-3 technical support across multiple customer facilities to ensure zero disruption to clinical operations.
- Design and develop RESTful APIs for OR medical devices, integrating hardware endpoints into the core application to enable seamless, centralized remote device management.
- Engineer internal tools using Ruby and Ansible to automate application OVA customer configurations and OR medical device provisioning, reducing manual setup times by 70%.

Software Application Developer/Analyst

May 2016 - October 2025

East End Hand Surgery

Port Jefferson, NY

- Engineered custom Python-based graphical user interfaces (GUIs) integrated with proprietary EMR systems to automate clinical workflows, saving healthcare providers 14+ hours weekly.
- Translated complex clinical business requirements into robust technical solutions by integrating and building custom applications for niche operational bottlenecks that significantly improved clinic throughput.
- Managed 2+ TB of sensitive patient records, ensuring data integrity, security, and adherence to HIPAA compliance through robust backup protocols and data management best practices.
- Authored comprehensive technical documentation and system configurations for all custom-built software.
- Served as the primary technical liaison between the clinical staff, EMR provider, and third-party vendors, proactively troubleshooting issues, coordinating system updates, and championing operational efficiency.
- Led the successful reorganization and modernization of the company's digital file structure, including the migration to an optimized, centralized storage solution, which reduced data storage and retrieval times by 50%.

Machine Learning Specialist - LLM Training & Debugging

April 2024 - October 2025

Data Annotation Tech

New York, NY

- Debugged and optimized code to effectively train and enhance large language model (LLM) responses.

Software Engineering & Data Science Fellow

July 2024 - May 2025

CUNY Tech Prep

New York, NY

- Selected for a competitive fellowship, applying SDLC principles to develop full-stack solutions and optimizing machine learning models via data engineering techniques (EDA, predictive modeling, & statistical evaluation).

PROJECTS

C-Sphere | [GitHub Repository](#)

May 2025

- Co-developed a full-stack web app and Chrome extension to summarize, organize, and search user bookmarks.
- Engineered a RAG backend pipeline combining content summarization, semantic vector search, and embedding-based storage in PostgreSQL for optimized data storage and intelligent content querying.

Breathe NYC | [GitHub Repository](#) | [Live Demo](#)

September 2024

- Developed a Python web application using NYC OpenData public datasets to visualize and predict seasonal AQI trends across NYC neighborhoods using a linear regression ML pipeline that achieved 97% accuracy.

MindfulNet AI | [GitHub Repository](#) | [Live Demo](#)

May 2024

- Engineered a full-stack Java application powered by a custom-built Random Forest machine learning model that predicts social media addiction with 98% accuracy to deliver personalized habit recommendations.

DropLock PDF | [GitHub Repository](#) | [Download Link](#)

July 2023

- Developed a Python-based Graphical User Interface (GUI) featuring drag-and-drop support for efficient, simultaneous encryption of multiple PDFs, enabling users to securely encrypt files in bulk and save time.

EDUCATION

Queens College, City University of New York

Master of Arts in Computer Science

May 2025

Bachelor of Arts in Computer Science

December 2024

Stony Brook University, State University of New York

Bachelor of Science in Biology

May 2019